REPRESENTABLE FUNCTORS ARE HOMOLOGICAL AND COHOMOLOGICAL

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 610-628, following the statement at 602-608.

Editorial note. No omissions: the filling argument is displayed; the contravariant half is the dual mirror.

Allowed standard prerequisites (editorial).

- Composition of consecutive triangle maps. lemma-composition-zero (same chapter, cited at ~613). g composed f = 0, the starting point of the LES.

Source reference. Original lemma 0146: sources/stacks-derived/pinned-original-source.tex, source lines 575–575. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a pre-triangulated category. For any object W of $\mathcal{D}$ the functor $\mathrm{Hom}_{\mathcal{D}}(W, -)$ is homological, and the functor $\mathrm{Hom}_{\mathcal{D}}(-, W)$ is cohomological.*

Proof. Consider a distinguished triangle (X, Y, Z, f, g, h). We have already seen that $g \circ f = 0$, see Lemma 0146. Suppose $a : W \to Y$ is a morphism such that $g \circ a = 0$. Then we get a commutative diagram

$$\begin{array}{ccccccc} W & \xrightarrow{1} & W & \longrightarrow & 0 & \longrightarrow & W[1] \\ \downarrow{\scriptstyle b} & & \downarrow{\scriptstyle a} & & \downarrow{\scriptstyle 0} & & \downarrow{\scriptstyle b[1]} \\ X & \longrightarrow & Y & \longrightarrow & Z & \longrightarrow & X[1] \end{array}$$

Both rows are distinguished triangles (use TR1 for the top row). Hence we can fill the dotted arrow b (first rotate using TR2, then apply TR3, and then rotate back). This proves the lemma. □

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.